



CENTER FOR INTEGRATED BUILDING SYSTEMS

College of Engineering, Architecture and Technology

CNN–based Prediction of Boiling Heat Transfer

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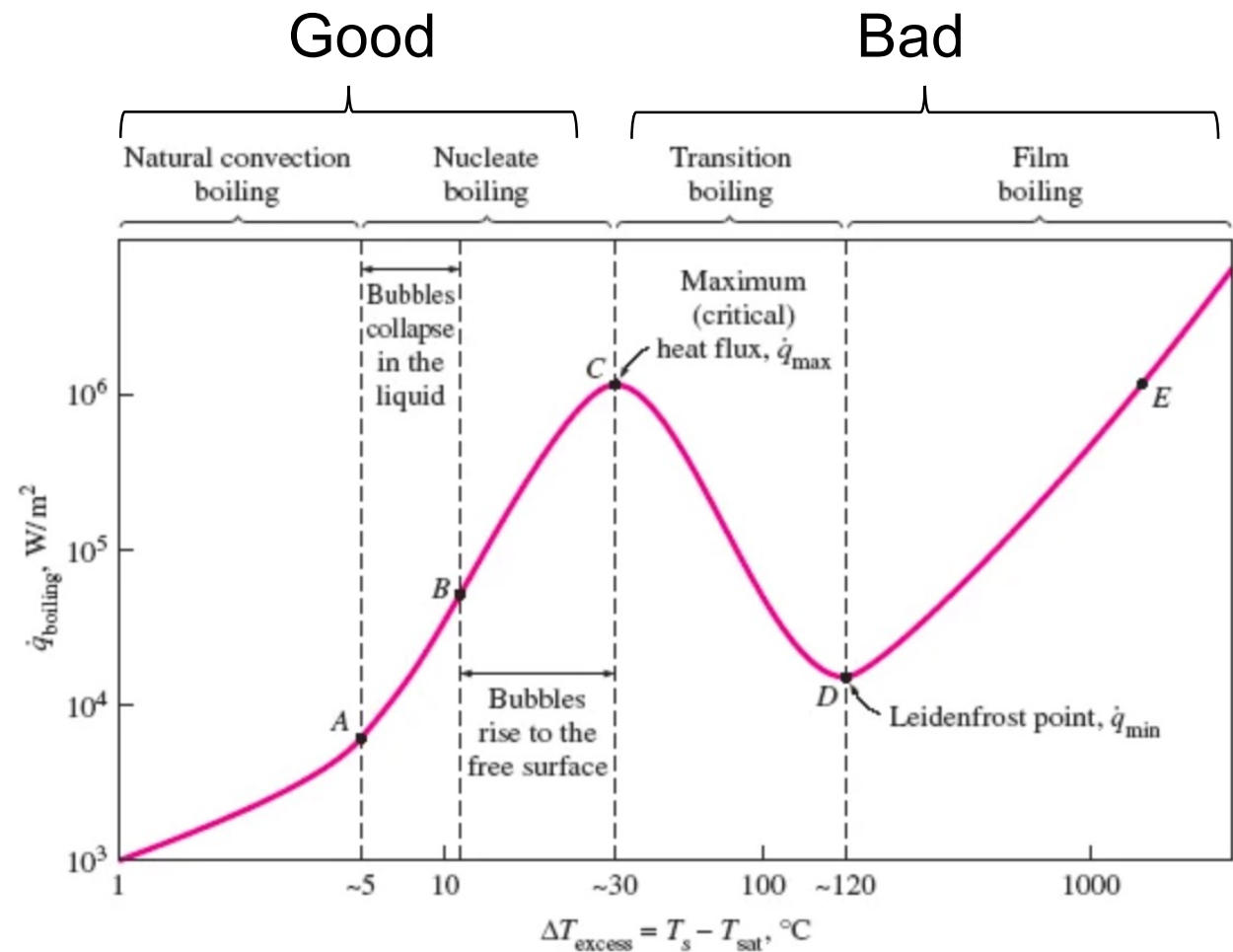
Introduction to Boiling

- Heat transfer is a limiting design factor in many high-energy systems
- Typical cooling strategies include forcing air/liquid over a heat source
 - » Convective heat transfer can carry energy away from heat sources
- Much more effective to boil a cooling fluid and take advantage of latent heat
 - » 10-100x better heat transfer



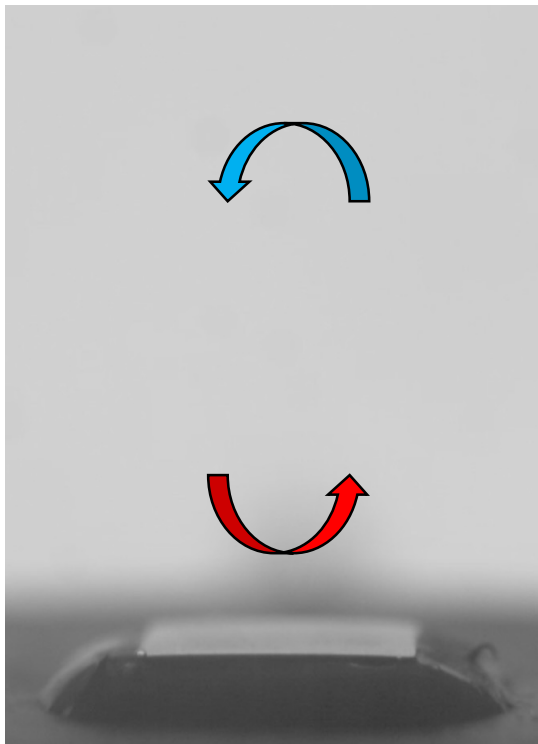
Introduction to Boiling

- Plot of boiling heat transfer
 - » Y-axis is heat transfer
 - » X-axis is temperature difference between fluid and surface
- Boiling takes place in different “regimes” based off conditions
 - » Early regimes are more stable and safer to operate in, but relatively low heat transfer
 - » Late regimes offer much higher heat transfer, but very unstable and unpredictable

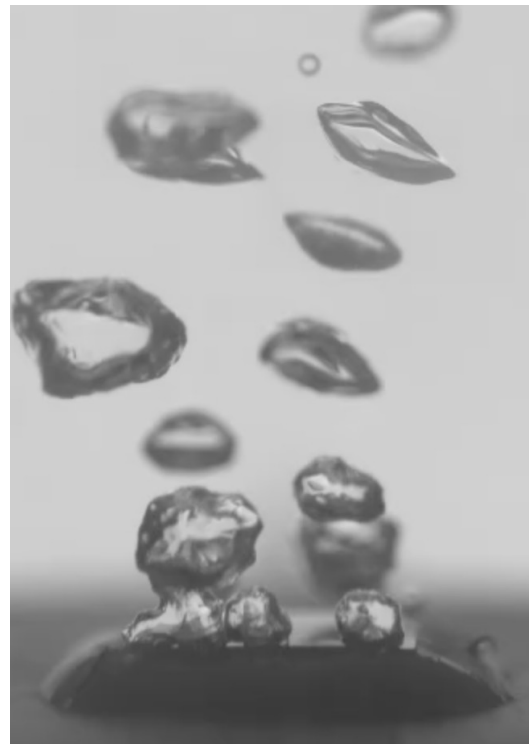


Introduction to Boiling

Natural
Convection
Boiling



Early Nucleate
Boiling



Late Nucleate
Boiling



Near CHF
Boiling



Introduction to Boiling

- Video shows unstable bubble growth and departure as boiling approaches critical heat flux
- Visually different to early boiling regimes
- Visual differences open opportunities to use CNNs



Target Metric for CNN

- As mentioned, it is beneficial to operate near CHF but important to understand instabilities to operate safely
- Existing CNN-based work exists to predict heat transfer and CHF, but we define a new metric: **CHF Proximity**

$$\Pi_{CHF} = \frac{T_{wall} - T_{sat}}{(T_{wall} - T_{sat})_{CHF}}$$

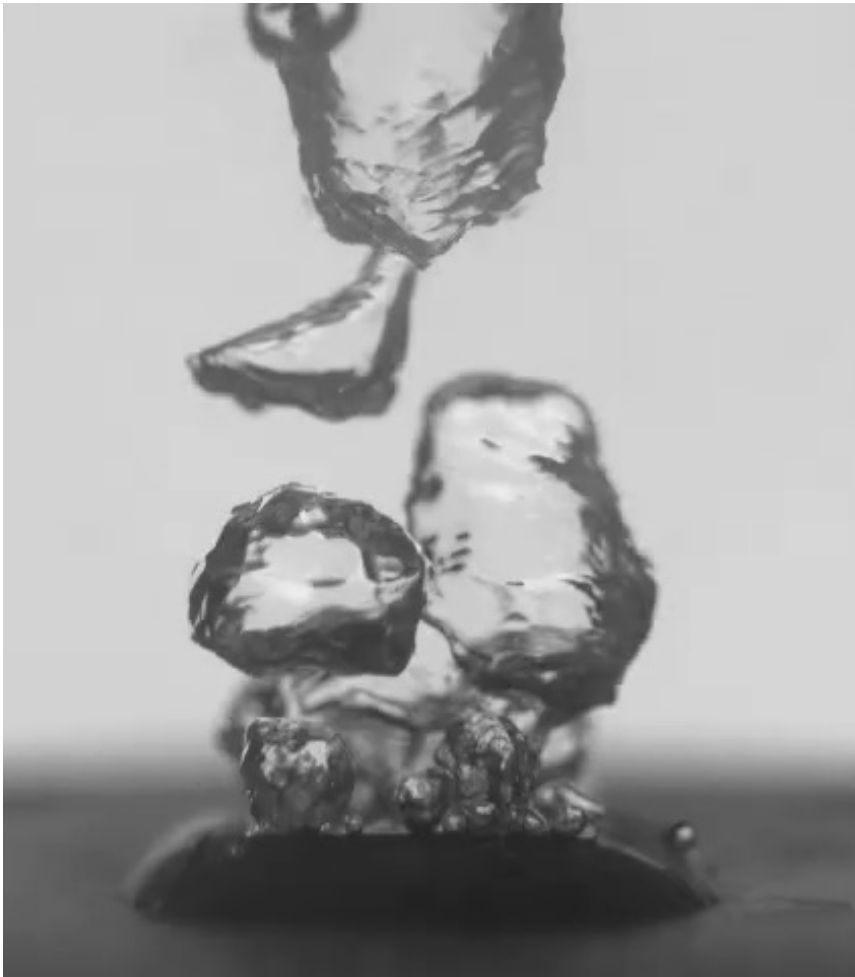
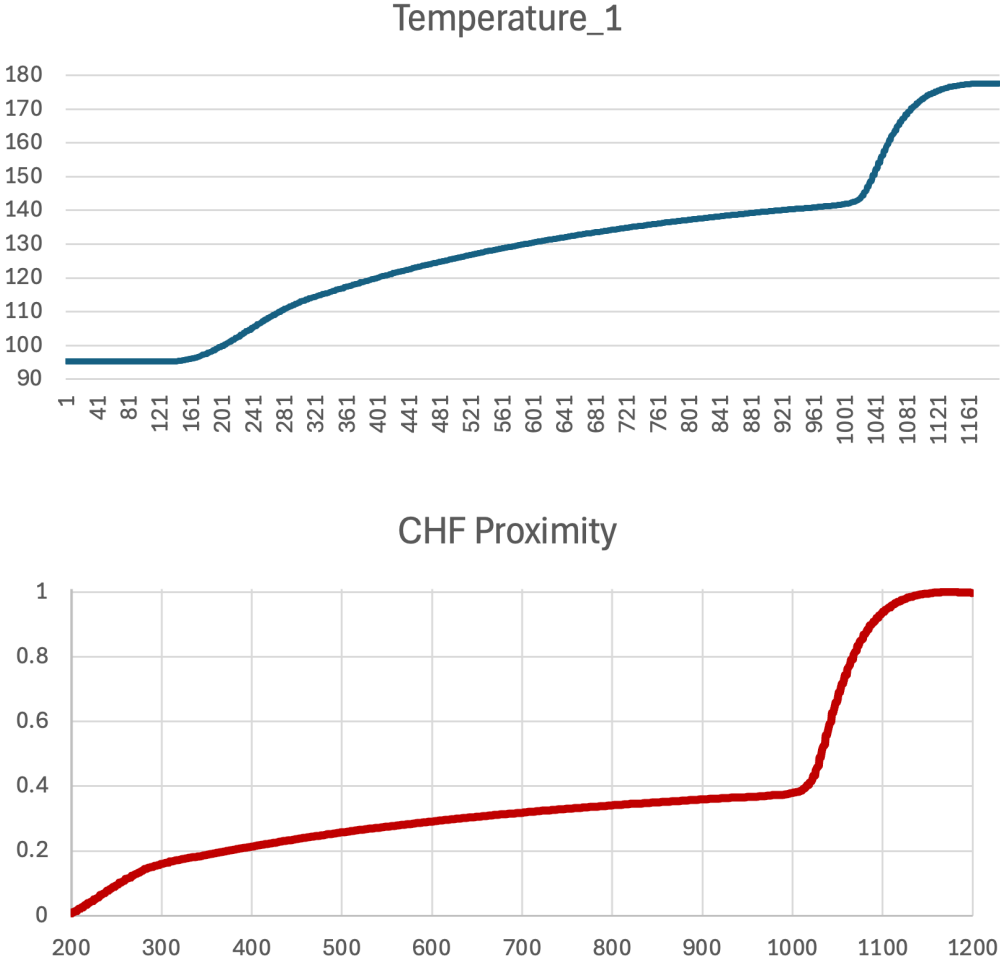
About the dataset

- Data from: “Unveiling hysteresis of transient boiling: A multimodal perspective”
 - » Published by Pandey et. al, in 2025
 - » Study to understand how boiling is different when ramping up and down in power using visual, audio, and temperature data
- Produced and published two full videos of boiling ramping up to CHF and back down, along with all temperature data
 - » Boiling_84: Deionized water on a copper foam surface
 - » Boiling_91: Deionized water on a smooth copper surface

Data Processing

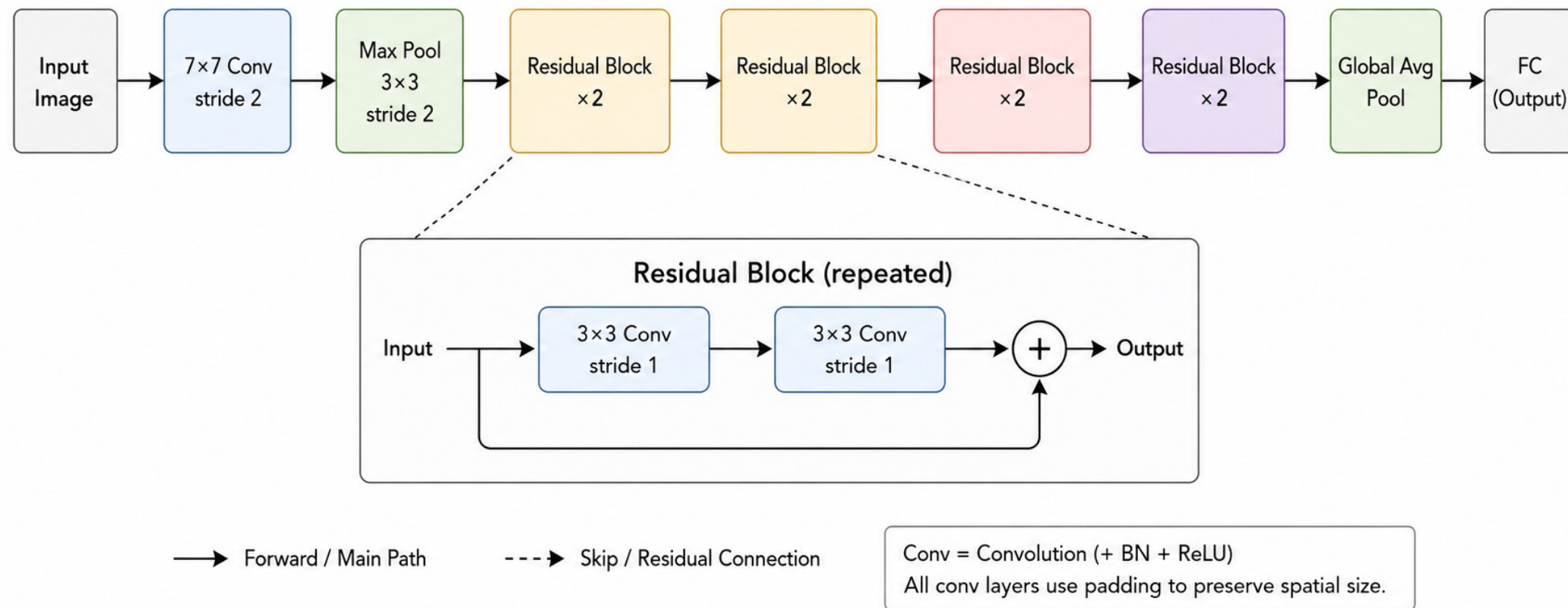
- Needed to create dataset of images and temperature points that are lined up
 - » Boiling_84: 45 minutes at 60fps, 1.5 million temperature points
 - » Boiling_91: 32 minutes at 60 fps, 1 million temperature points
- Original dataset was far too data dense, so down-sampling was done
- Python script created to chop up raw data and created linked image frame at every temperature point
- Resulting in 1200 data points from each set from zero boiling up to CHF
- One set for training, one set for testing

Data Processing



CNN Network Choice

- Decided to go with ResNet-18 (reasoning shown in next slides)



CNN Network Choice

- Dataset is relatively small, likely insufficient to train a custom model
- Chose to use a pretrained model, perform training on pretrained weights

- » AlexNet
- » ResNet
- » InceptionNet
- » Etc.

- Classification of boiling regimes, fluids, and heating surfaces through deep learning algorithms and image analysis
 - Published by Paz et. al, in 2025



CNN Architecture	Model F1 Score
Basic CNN	91.11%
AlexNet	98.87%
ResNet	96.50%
InceptionNet	94.23%

Testing different networks

- Results of my own network experimentation:

CNN Architecture	Best MAE (%)	RMSE (%)	R^2	Correlation
AlexNet	13.5	23.6	0.429	0.789
ResNet-18	11.3	21.9	0.505	0.792
ResNet-34	13.9	24.4	0.388	0.737
EfficientNet	12.6	21.7	0.514	0.796

Tuning ResNet-18 Model

Different ways I tried to improve the model:

1. Training Hyperparameter Tuning:

- Batch size optimal at 32
- Learning rate best at $1e-4$
- Weight Decay at $1e-4$
- Backbone not frozen during training

14.7% MAE



11.3% MAE

2. Data Augmentation

- Color Jitter made very small improvement
- Random scaling and translation made larger impact

3. Output layer dropout

- Dropout = 0.2 was beneficial, 0.4 was too much



Final Model and Results

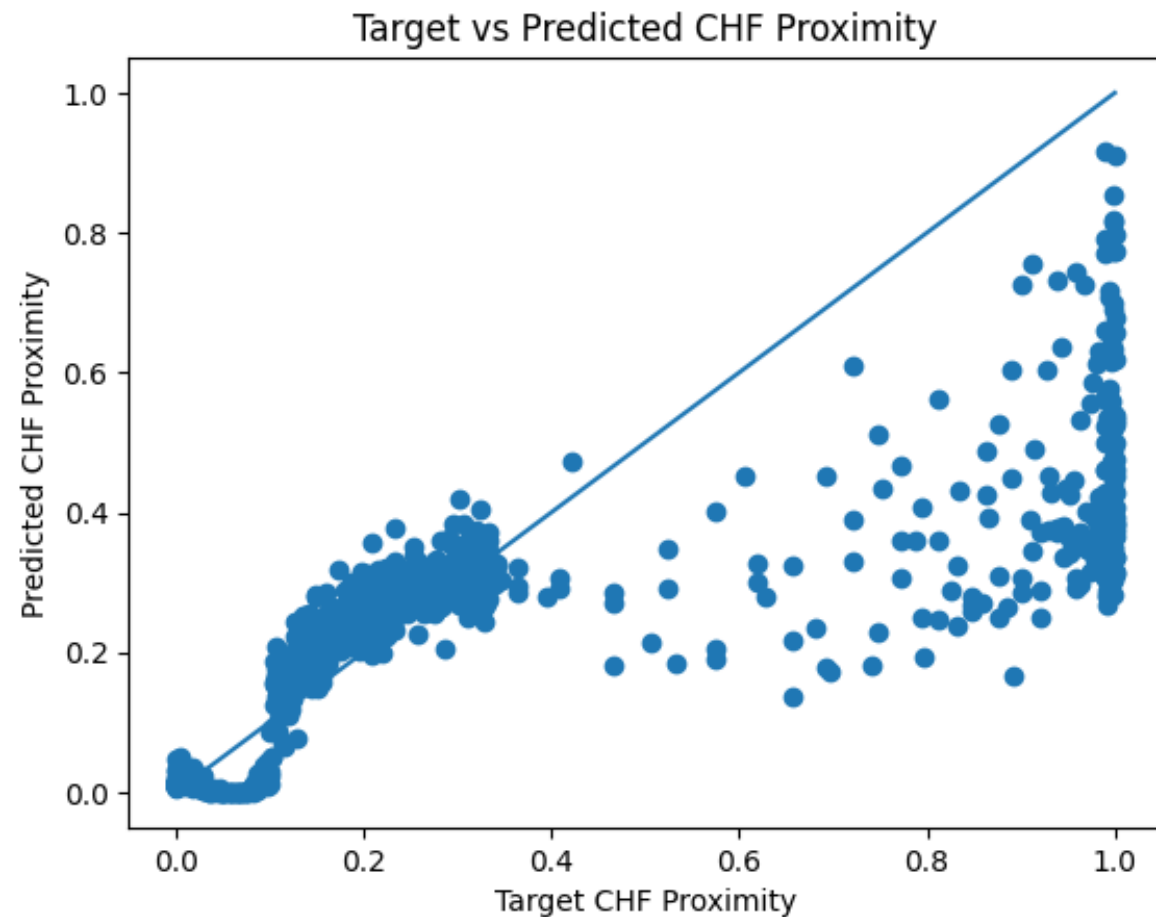
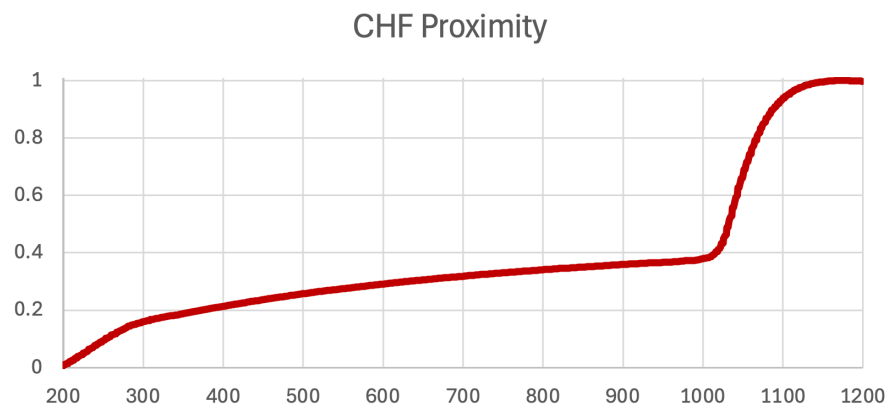
Final Model:

- ResNet-18 with Sequential Output Layer
 - One feedforward layer
 - Dropout: 0.2
- Epochs: 10
- Learning Rate: 1e-4
- Weight Decay: 1e-4
- Validation Split: 0.2

Performance Metric	
MAE (%)	11.3
RMSE (%)	21.9
R^2	0.505
Correlation	0.792

Final Model and Results

- Target vs Predicted for ResNet-18
- Shows very strong prediction for targets between 0.1 to 0.4
- Very poor performance above 0.5 due to lack of data



Conclusions and Future Work

- ResNet-18 outperformed other tested networks
 - Hyperparameters tuned to optimize performance
 - Data augmentation improved generalization
 - Dropout improves generalization
- Shortage of training data
 - Model was bound to struggle with 50/50 train/test split
- Need to account for data imbalance
- Investigate dynamic CNN models
 - Leverage time series stream of images

References

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- Paz, C., Cabarcos, A., Concheiro, M., & Conde-Fontenla, M. (2025). Classification of boiling regimes, fluids, and heating surfaces through deep learning algorithms and image analysis. *International Journal of Heat and Mass Transfer*, 242, 126829. <https://doi.org/10.1016/j.ijheatmasstransfer.2025.126829>

Thank you for listening!

